

BLACK RIVER FALLS AREA, WI, USA

WMO: 720859

Lat: 44.251N

Lon: 90.855W

Elev: 255

StdP: 98.30

Time Zone: -6.00 (NAC)

Period: 10-19

WBAN: 00286

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.2	(c) -25.1	(d) -33.3	(e) 0.2	(f) -29.0	(g) -29.1	(h) 0.3	(i) -24.5	(j) 8.3	(k) -8.7	(l) 7.5	(m) -8.0	(n) 0.3	(o) 170	(p) 0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.7	(c) 32.5	(d) 22.4	(e) 30.9	(f) 21.4	(g) 29.2	(h) 20.4	(i) 24.4	(j) 29.2	(k) 23.3	(l) 28.0	(m) 22.4	(n) 26.8	(o) 3.7	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.3	26.0	22.1	17.3	25.0	21.1	16.2	24.3	75.1	29.1	70.7	28.7	67.1	26.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.7	6.8	5.6	-33.8	34.8	2.5	1.8	-35.6	36.1	-37.1	37.2	-38.5	38.2	-40.4	39.5
			-34.0	26.1	2.5	1.2	-35.8	27.0	-37.2	27.7	-38.6	28.4	-40.5	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-9.3	-8.0	-0.4	6.3	13.9	18.9	21.2	19.4	15.6	8.0	0.6	-5.6
	DBStd	12.05	7.80	7.61	7.29	5.05	5.05	3.54	3.26	3.24	4.94	4.94	5.76	6.83
	HDD10.0	2457	599	503	330	131	18	0	0	0	8	99	285	483
	HDD18.3	4467	857	737	580	361	154	36	10	26	111	322	532	742
	CDD10.0	1290	0	0	9	21	141	266	347	290	177	37	3	0
	CDD18.3	257	0	0	1	0	18	51	99	57	30	1	0	0
	CDH23.3	3390	0	0	7	22	312	669	1269	739	359	12	0	0
	CDH26.7	1097	0	0	1	2	93	209	474	208	109	1	0	0
Wind	WSAvg	2.1	2.2	2.4	2.4	2.7	2.3	1.9	1.6	1.5	1.7	2.1	2.2	2.0
Precipitation	PrecAvg	906	31	25	47	83	107	140	109	122	95	69	42	32
	PrecMax	1115	74	51	87	168	268	261	267	271	276	159	82	65
	PrecMin	658	10	9	13	27	40	53	25	26	21	24	2	9
	PrecStd	107	16	12	23	45	49	64	58	63	59	37	23	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	14.0	23.6	26.3	32.5	33.3	35.9	33.0	32.7	24.8	19.3	10.7
		MCWB	4.0	7.7	16.4	14.8	19.2	23.2	23.7	22.7	21.9	15.6	12.3	7.0
	2%	DB	5.1	7.9	18.4	22.4	29.1	31.2	32.7	30.8	29.9	22.7	15.7	7.6
		MCWB	3.0	4.4	12.2	12.1	17.4	20.7	23.5	21.5	20.7	15.2	10.8	5.6
	5%	DB	2.8	5.1	14.7	19.7	26.8	29.1	30.9	28.9	27.4	20.2	12.5	5.4
		MCWB	1.3	2.4	9.5	10.6	16.5	19.7	22.3	20.9	19.5	14.9	8.2	3.6
	10%	DB	1.3	2.5	10.7	16.7	24.5	27.3	29.3	27.5	25.1	17.5	9.8	2.9
		MCWB	0.4	0.4	7.0	8.9	16.0	19.1	21.0	19.8	18.8	12.7	7.0	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	8.1	17.0	16.4	21.8	24.9	26.8	25.0	23.9	20.7	14.1	8.8
		MCDB	7.2	12.3	22.9	21.0	27.7	30.8	31.9	28.0	28.3	22.1	16.3	9.1
	2%	WB	3.0	5.2	14.7	13.8	20.1	23.1	25.1	23.7	22.6	17.8	11.6	6.0
		MCDB	4.8	7.6	17.9	19.7	25.1	28.0	29.7	27.6	26.6	19.6	16.0	6.5
	5%	WB	1.7	2.5	9.8	12.2	18.9	21.9	23.8	22.6	21.6	15.6	8.8	3.3
		MCDB	2.6	5.1	13.0	16.9	24.0	26.2	28.9	26.9	24.9	18.3	12.0	5.0
	10%	WB	0.4	0.6	6.5	10.1	17.5	20.5	22.6	21.5	20.3	13.6	6.8	1.9
		MCDB	1.3	2.2	10.2	16.2	21.9	24.2	27.3	25.4	23.8	16.9	9.2	2.9
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.8	13.0	14.1	14.8	14.3	14.7	14.7	15.1	13.2	10.9	9.4
		MCDBR	11.7	14.3	17.2	20.4	18.6	16.9	15.8	15.7	16.5	16.9	16.5	11.5
		MCWBR	9.3	10.7	10.3	10.8	8.4	8.1	7.1	7.5	8.1	10.4	11.3	9.2
	5% WB	MCDBR	10.9	13.3	13.6	16.0	14.4	13.1	13.6	12.8	12.6	12.6	14.5	9.6
		MCWBR	8.9	10.3	8.7	9.6	7.1	7.4	6.9	6.9	7.3	9.7	10.5	8.0
Clear-Sky Solar Irradiance	taub	0.258	0.286	0.320	0.360	0.392	0.398	0.392	0.397	0.362	0.327	0.293	0.265	
	taud	2.404	2.323	2.383	2.345	2.300	2.318	2.345	2.332	2.424	2.508	2.518	2.455	
	Ebn at Noon	881	907	919	902	879	872	873	855	862	853	834	838	
	Edh at Noon	73	96	104	118	127	126	121	118	100	80	67	64	
All-Sky Solar Radiation	RadAvg	1.76	2.87	3.94	4.52	5.37	5.93	6.16	5.21	4.15	2.65	1.71	1.28	
	RadStd	0.19	0.27	0.46	0.44	0.41	0.46	0.29	0.35	0.30	0.34	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page